

BITWISE OPERATORS

Bitwise operator's works on bits.

Turbo-c is a 16 bit compiler. Due to this bitwise operations are limited to 16 bits only [2^0 to 2^{15}].

Bitwise operators operate **integer** type values only.

We have to calculate only the **on** bits [**1**].

When the first bit[**Sign bit**] is **1** then the number is **Negative** and it is **0** then the number is **positive**.

They are very much used in system software development.

Note: Bitwise operator is low level feature.

C-Language supports following bitwise operators.

& -Bitwise and

| - Bitwise or

^ - XOR ==> Exclusive OR

~ - Compliment operator

<< - Left shift operator

>> - Right shift operator

& - Bitwise and: In this both bits are 1's then result bit is 1. Otherwise result bit is 0.

Eg: **25 & 15 = 9**

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111

$$\begin{array}{r} 2 \overline{) 25} \\ 2 \overline{) 12 - 1} \\ 2 \overline{) 6 - 0} \\ 2 \overline{) 3 - 0} \\ 1 - 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 15} \\ 2 \overline{) 7 - 1} \\ 2 \overline{) 3 - 1} \\ 1 - 1 \end{array}$$

25 & 15 = 9

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111

&

$$\begin{array}{r} \hline 0000 \ 0000 \ 0000 \ 1001 \\ \downarrow \quad \downarrow \\ 2^3 + 2^0 \\ \downarrow \quad \downarrow \\ 8 + 1 = 9 \end{array}$$

| - Bitwise or: In this both bits are 0's then result bit is 0. Otherwise result bit is 1.

Eg: $25 \mid 15 = 31$

$$25 \mid 15 = 31$$

25 = 0000 0000 0001 1001

15 = 0000 0000 0000 1111

0000 0000 0001 1111

$2^4 + 2^3 + 2^2 + 2^1 + 2^0$

$16 + 8 + 4 + 2 + 1 = 31$

^ - XOR [Exclusive OR]: In this both bits are same then result bit is 0. Otherwise result bit is 1.

Eg: $25 \wedge 15 = 22$

$$25 \wedge 15 = 22$$

25 = 0000 0000 0001 1001

15 = 0000 0000 0000 1111

$\wedge$

0000 0000 0001 0110

$$\begin{array}{r}
 2^4 + 2^2 + 2^1 \\
 16 + 4 + 2 = 22
 \end{array}$$

~ - Complement operator: In complement operation the bits are complimented. i.e. 1's become 0's and 0's become 1's. Due to this +Ve no becomes -Ve and -Ve no becomes +Ve.

eg: ~25 → -26

25 = 0000 0000 0001 1001
1111 1111 1110 0110
-128+64+32+4+2=-26
-128 + 102 = -26

$$25 = 0000\ 0000\ 0001\ 1001$$

$$\sim = 1111\ 1111\ 1110\ 0110$$

$$\begin{array}{cc} \diagdown & \diagup \\ 5 & 2 \end{array}$$

$$2+4+32+64+128+256+512+1024+2048+4096+8192+16384-32768=-26$$

$$\sim -25 = 0000\ 0000\ 0001\ 1001$$

$$1's\ \sim = 1111\ 1111\ 1110\ 0110$$

$$2's\ \sim = \begin{array}{r} 0000\ 0000\ 0000\ 0001 \\ 1111\ 1111\ 1110\ 0111 \end{array}$$

$$\begin{array}{cc} \diagup & \diagdown \\ 24 & 2 \end{array}$$

$$16+8=24$$

$$\begin{array}{cccc} 1 & 0 & 0 & 01 \\ 0 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 10 \end{array}$$

Note: When starting bit is 1 given no is –Ve.

Eg: $\sim -25 \rightarrow +24$

$$\sim -25 = +24$$

$$25 = \begin{array}{|c|c|} \hline 0000 & 0000 \\ \hline \end{array} \begin{array}{c} 0001\ 1001 \\ 1111\ 1111\ 1110\ 0110 \end{array}$$

$$\begin{array}{|c|c|} \hline 1111 & 1111 \\ \hline \end{array} \begin{array}{c} 1110\ 0110 \\ +1 \end{array}$$

$\Leftarrow$ 1's compliment

$\Leftarrow$ 2's Complement

$$\begin{array}{cccc} 1111 & 1111 & 1110 & 0111 \\ \sim & \rightarrow & 0000 & 0000 & 0001 & 1000 \end{array}$$

$$\downarrow \downarrow$$

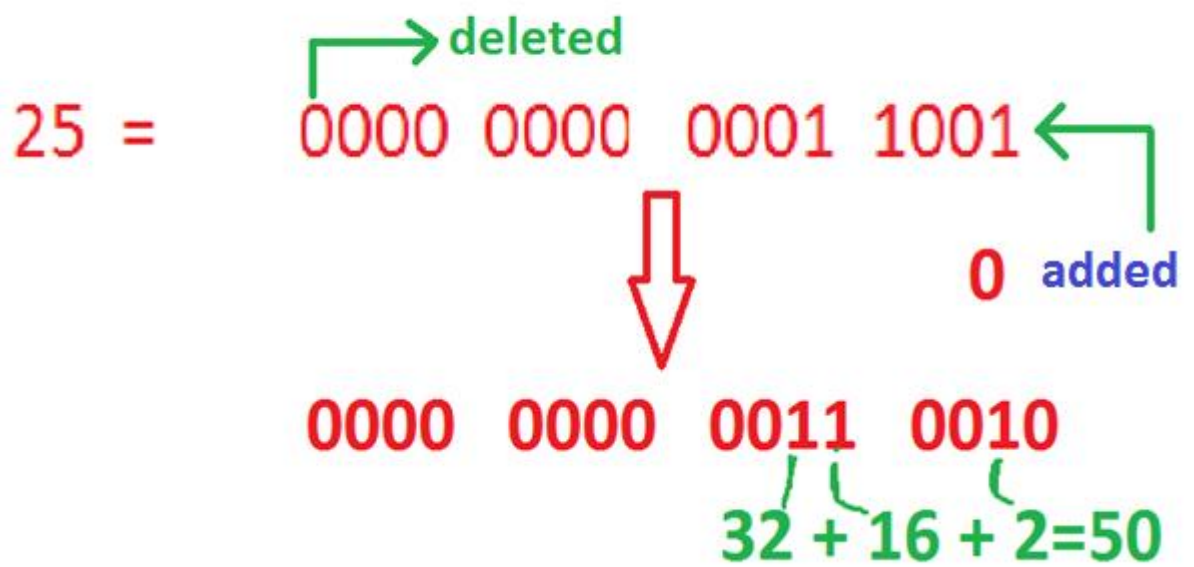
$$16+8=24$$

<< - left shift operator:

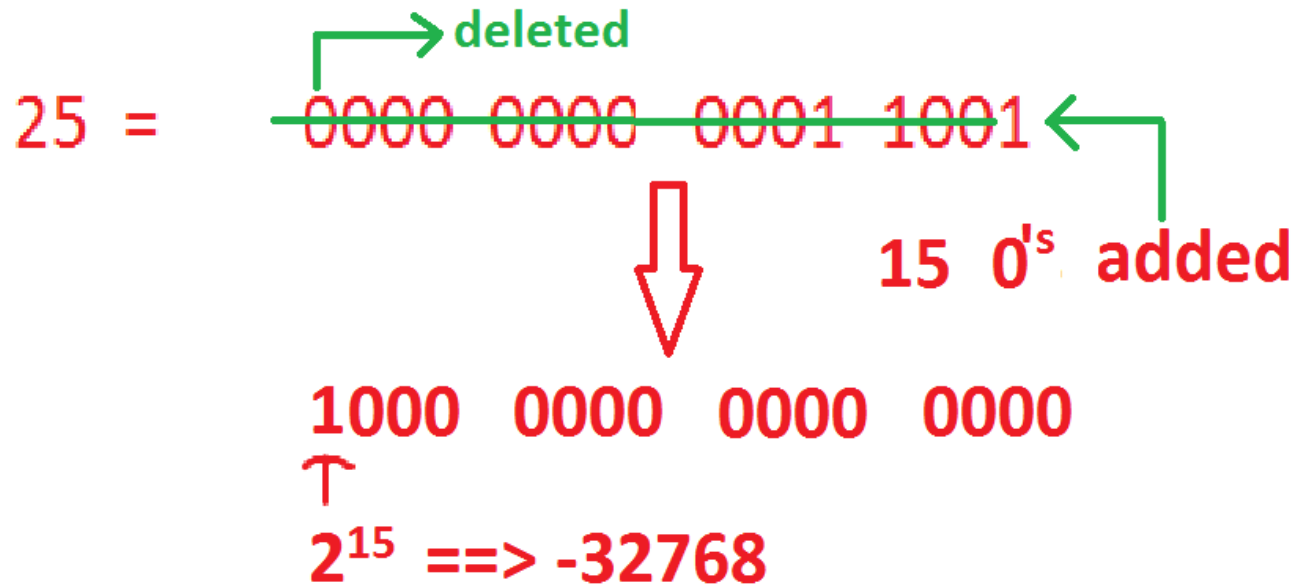
In left shift operation, the specified no of bits are deleted from left side and the same no of **zeros** added on right side. In left shift operation, most probably the value is multiplied with 2 that no of times.

Eg: $25 \ll 1 = 50$, $25 \ll 2 = 100$, $25 \ll 15 = -32768$, $25 \ll 16 = 0$

eg: $25 \ll 1 = 50$



eg: $25 \ll 15 = -32768$



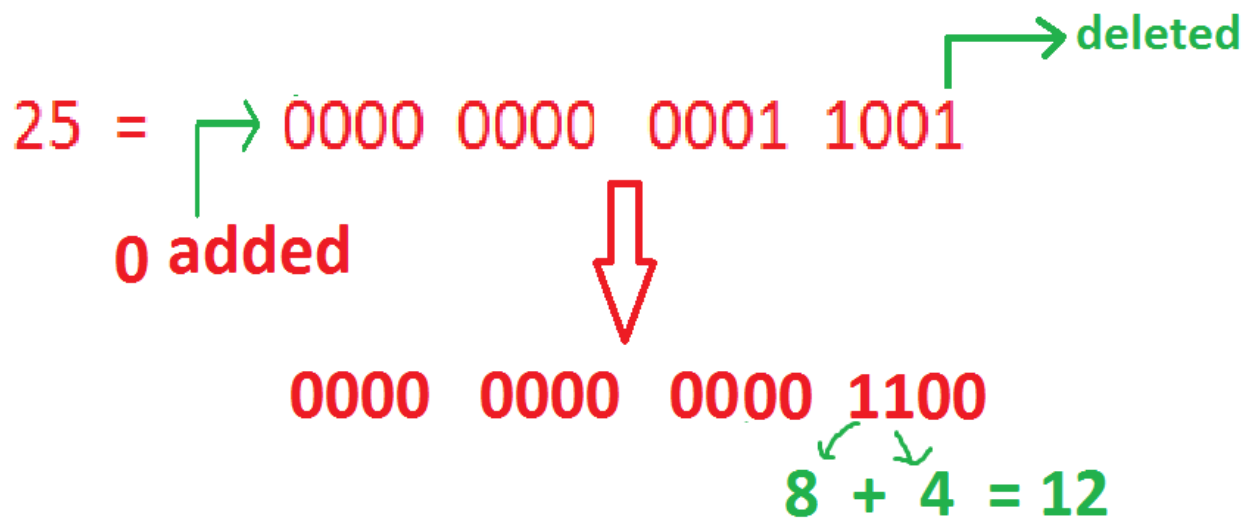
Note: When starting bit 1 no is negative.

>> - Right shift operator:

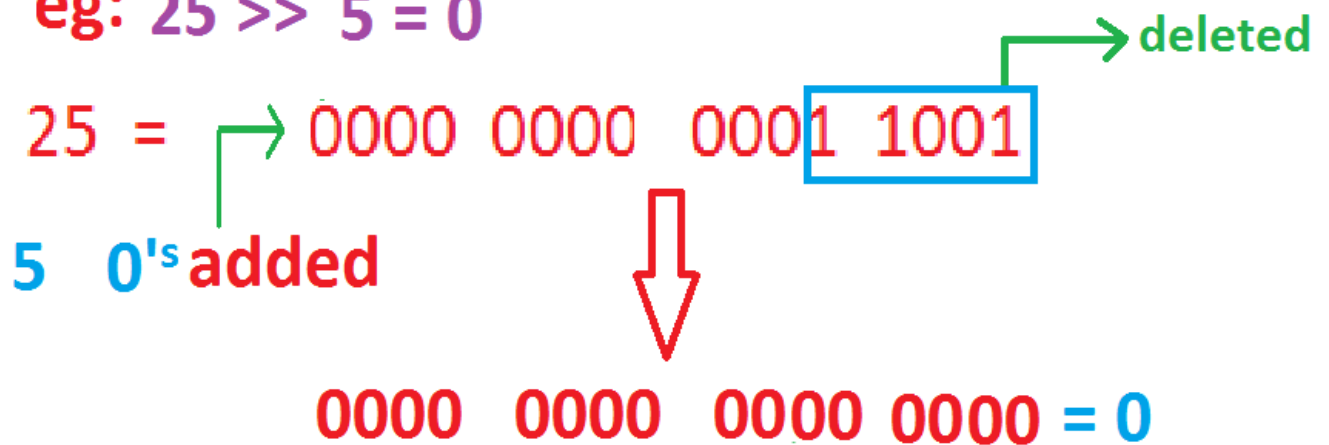
In right shift operation, the bits are moved to right side i.e. the specified no. of bits are deleted from right side and same no. of **zero's** are added left side. Due to this always the number is divided with 2 that no of times.

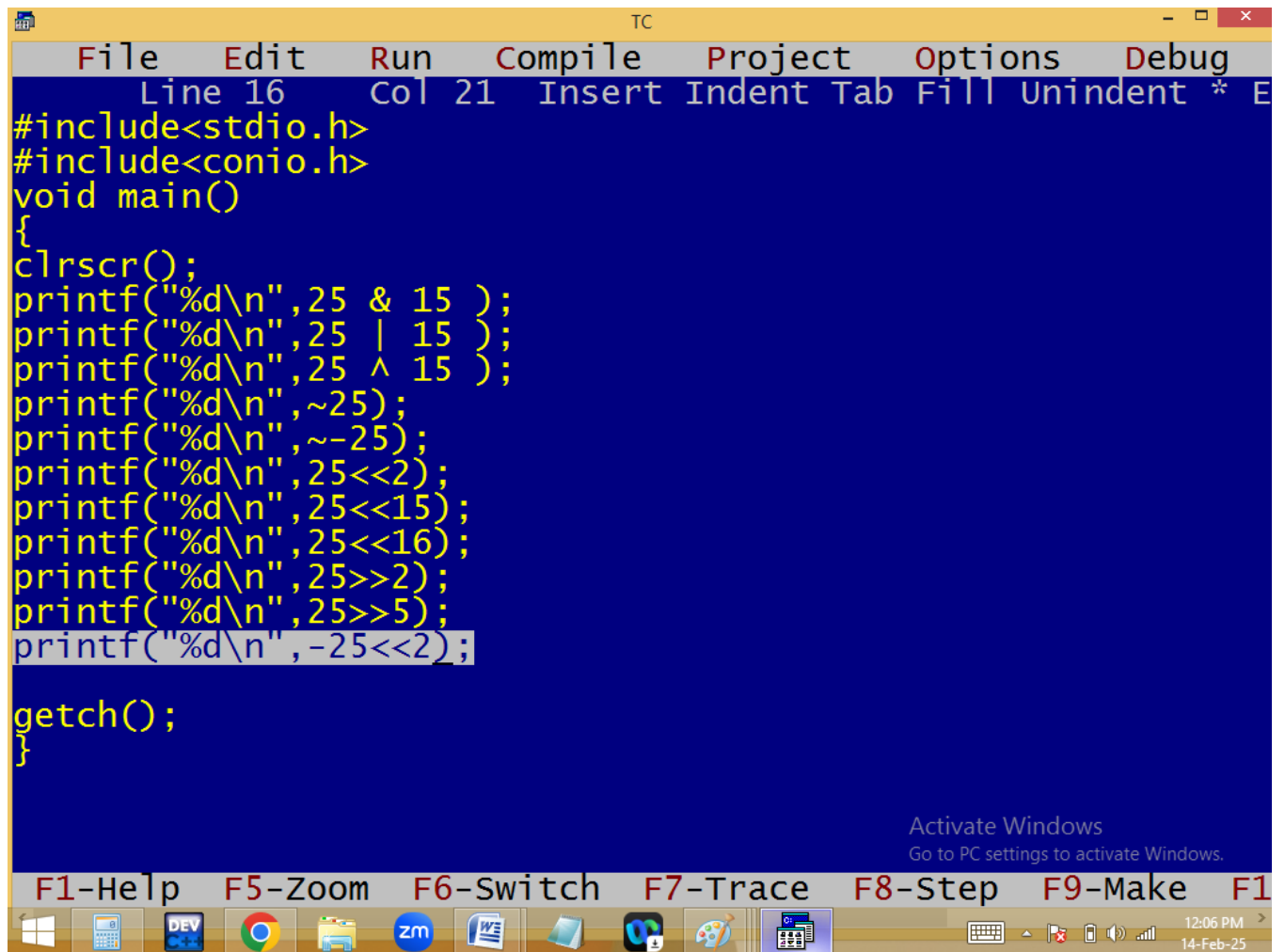
Eg: $25 \gg 1 = 12$, $25 \gg 2 = 6$, $25 \gg 3 = 3$, $25 \gg 4 = 1$, $25 \gg 5 = 0$

eg: $25 \gg 1 = 12$



eg: $25 \gg 5 = 0$





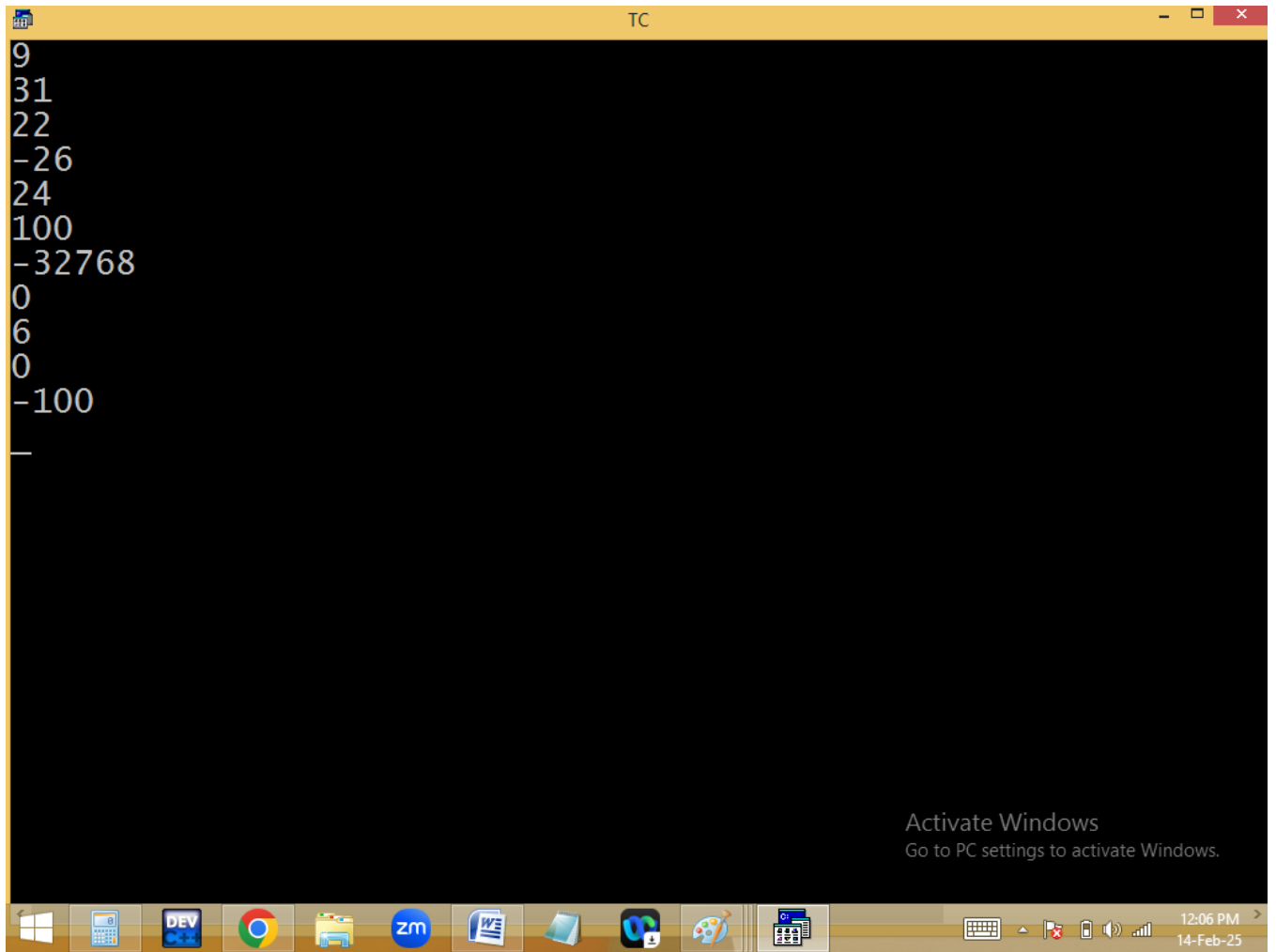
```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 21 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",25 & 15 );
printf("%d\n",25 | 15 );
printf("%d\n",25 ^ 15 );
printf("%d\n",~25);
printf("%d\n",~-25);
printf("%d\n",25<<2);
printf("%d\n",25<<15);
printf("%d\n",25<<16);
printf("%d\n",25>>2);
printf("%d\n",25>>5);
printf("%d\n",-25<<2);

getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

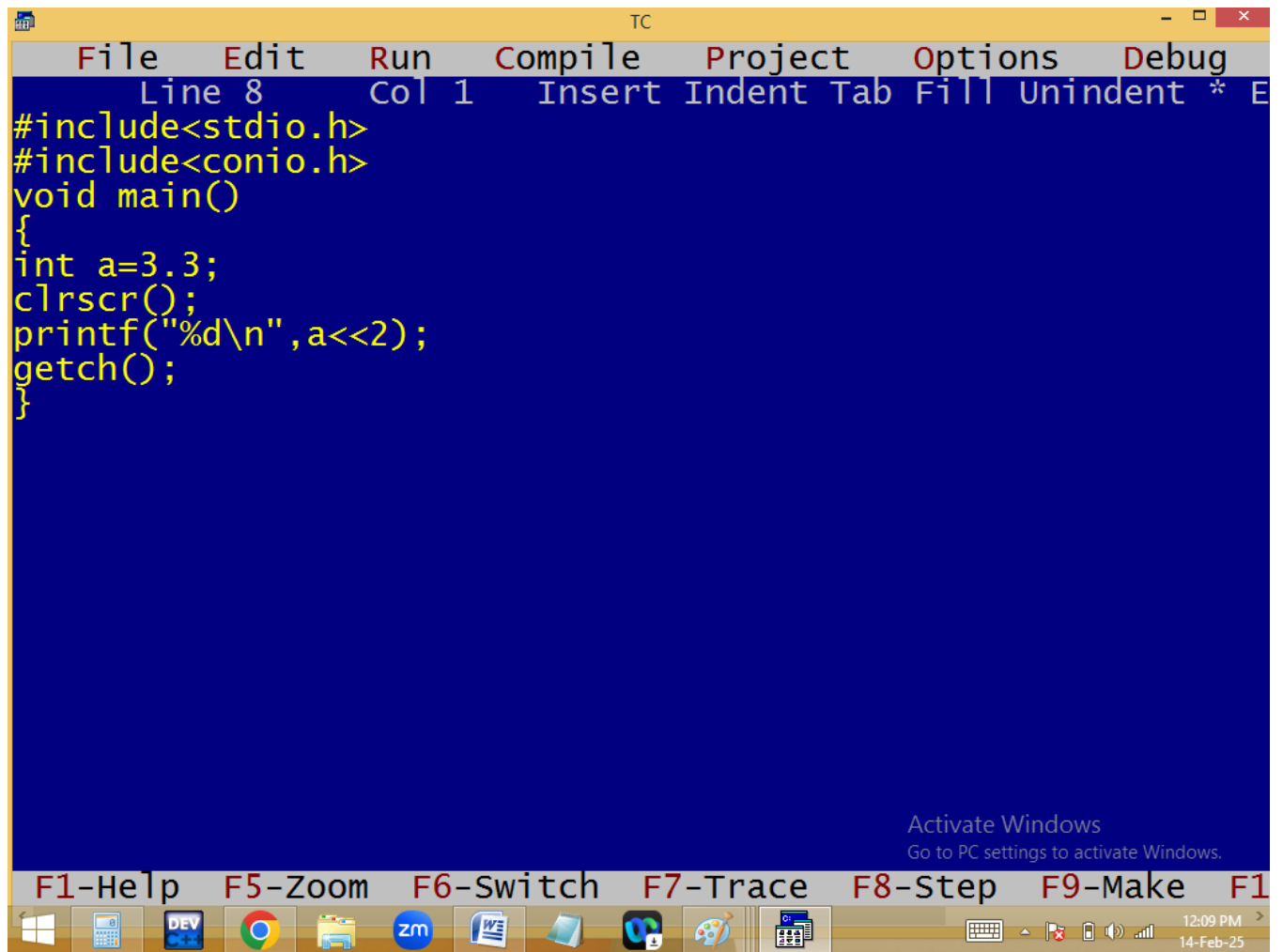
12:06 PM
14-Feb-25



```
9
31
22
-26
24
100
-32768
0
6
0
-100
—
```

Activate Windows
Go to PC settings to activate Windows.

12:06 PM
14-Feb-25

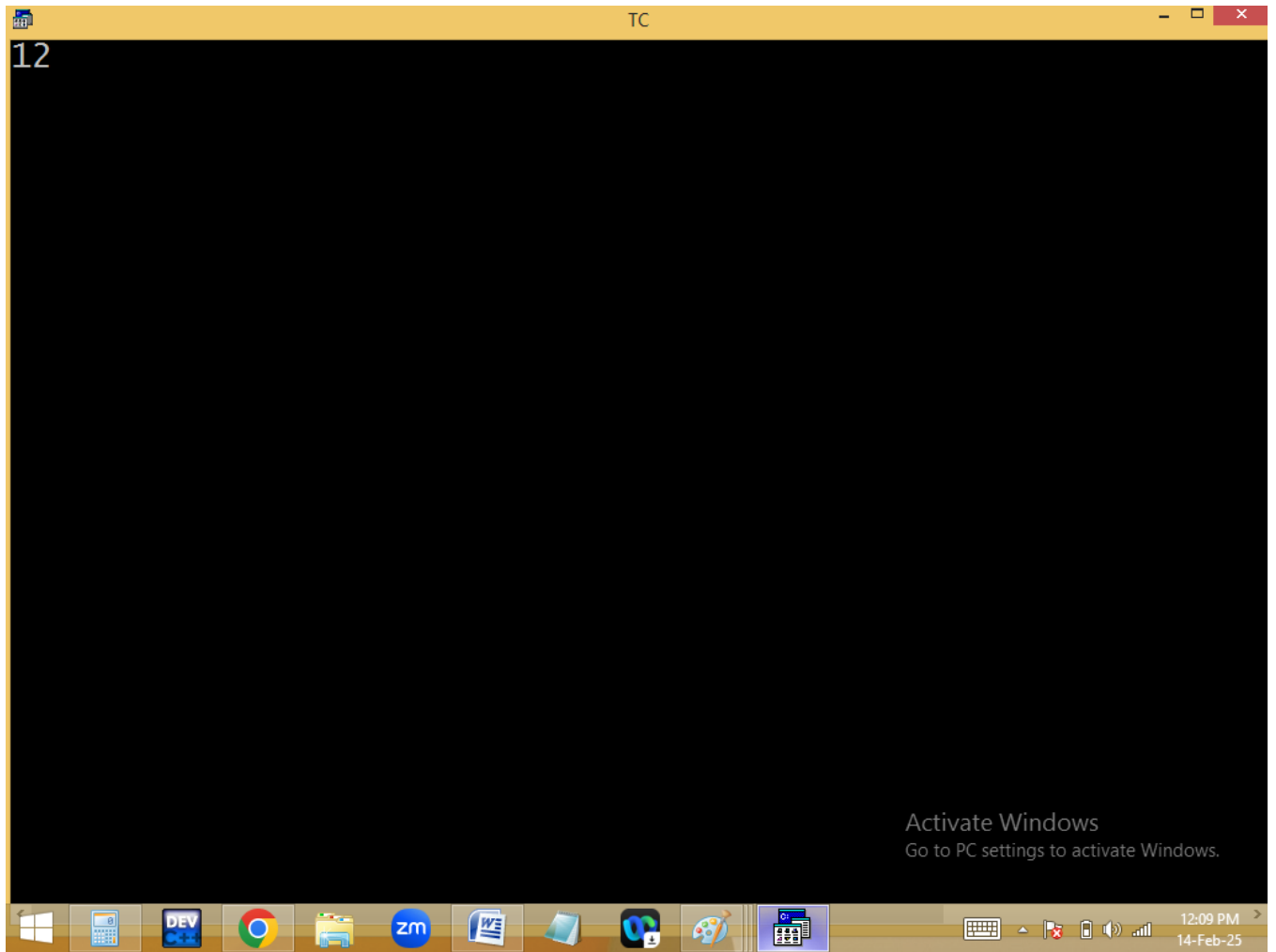


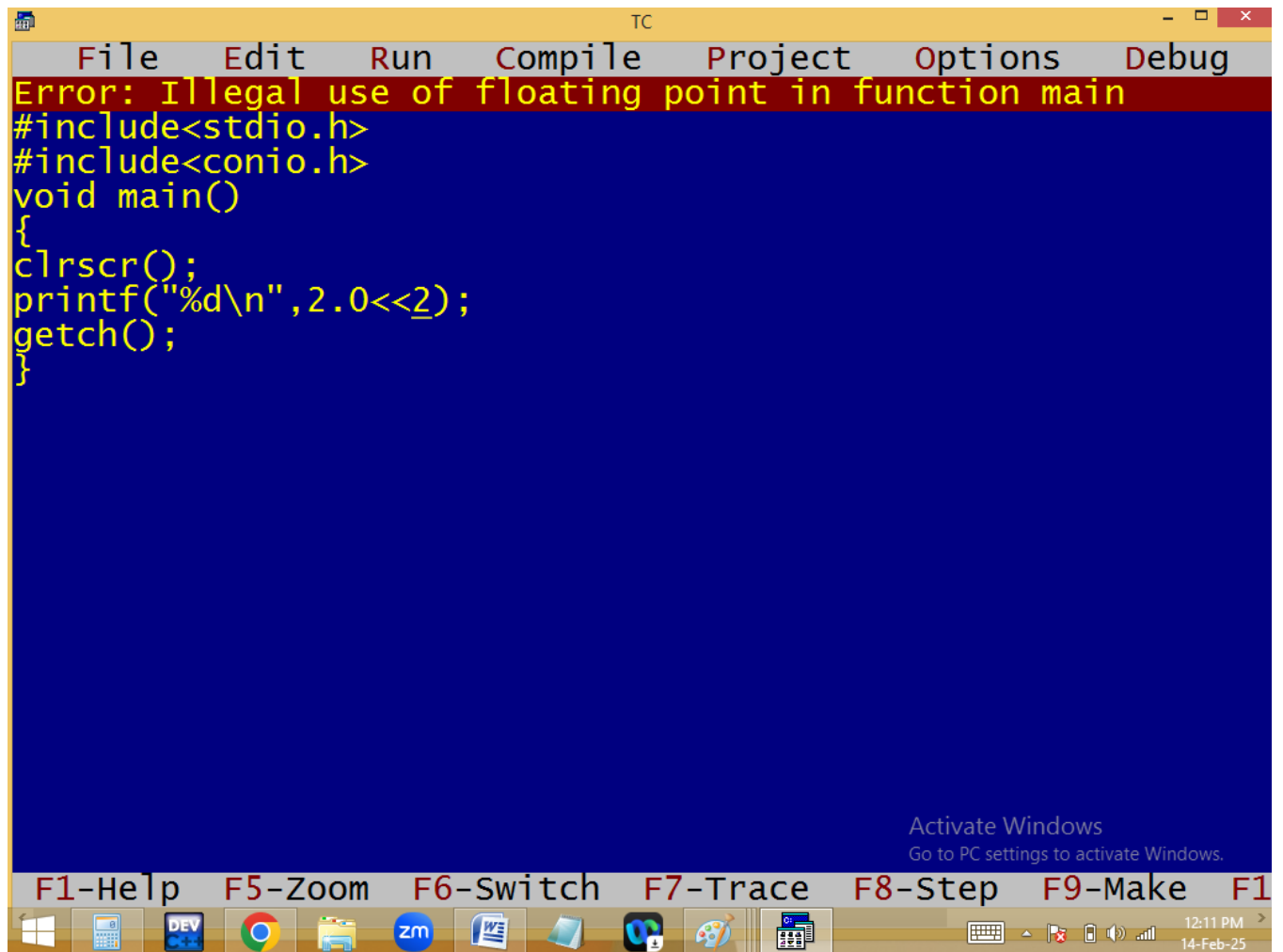
```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3.3;
clrscr();
printf("%d\n",a<<2);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

12:09 PM
14-Feb-25

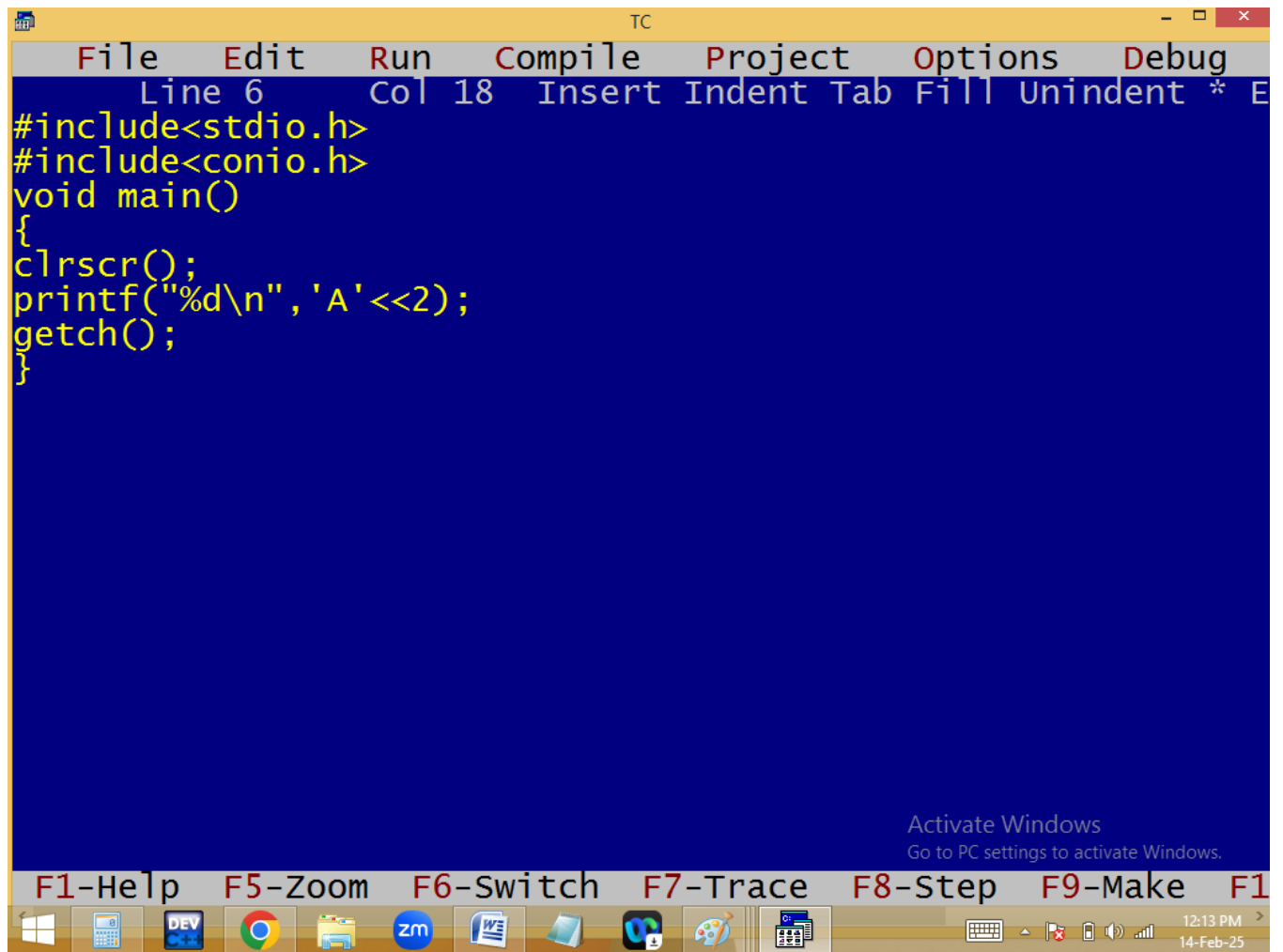




The screenshot shows the Turbo C++ (TC) IDE interface. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top states: "Error: Illegal use of floating point in function main". Below this, the source code is displayed on a blue background with yellow text:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",2.0<<2);
getch();
}
```

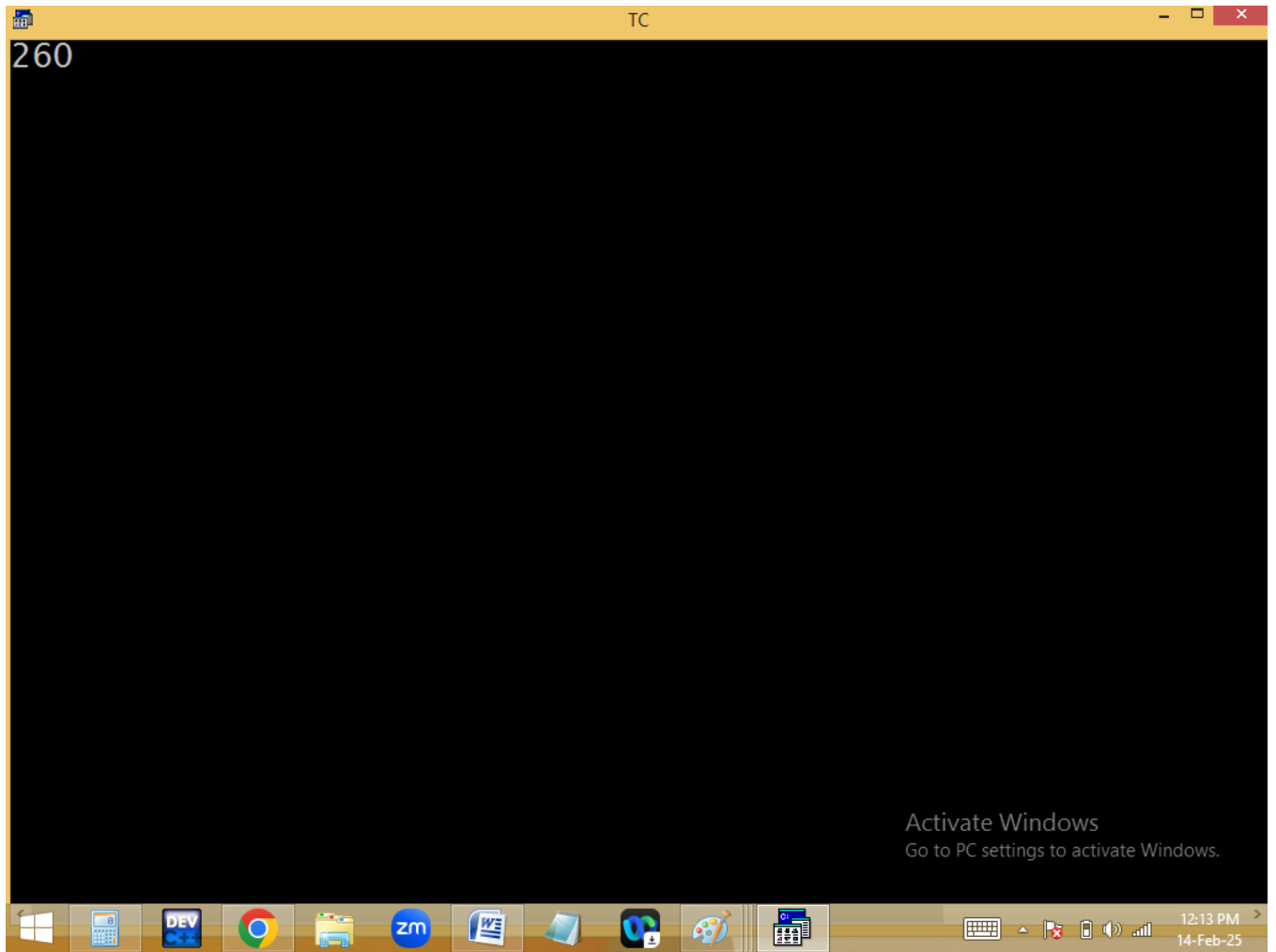
At the bottom of the IDE window, a status bar contains function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, and several applications (DEV, Chrome, File Explorer, Zoom, Word, PowerPoint, Paint, and a calculator). The system tray on the right shows the date and time: "12:11 PM" and "14-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the desktop area.

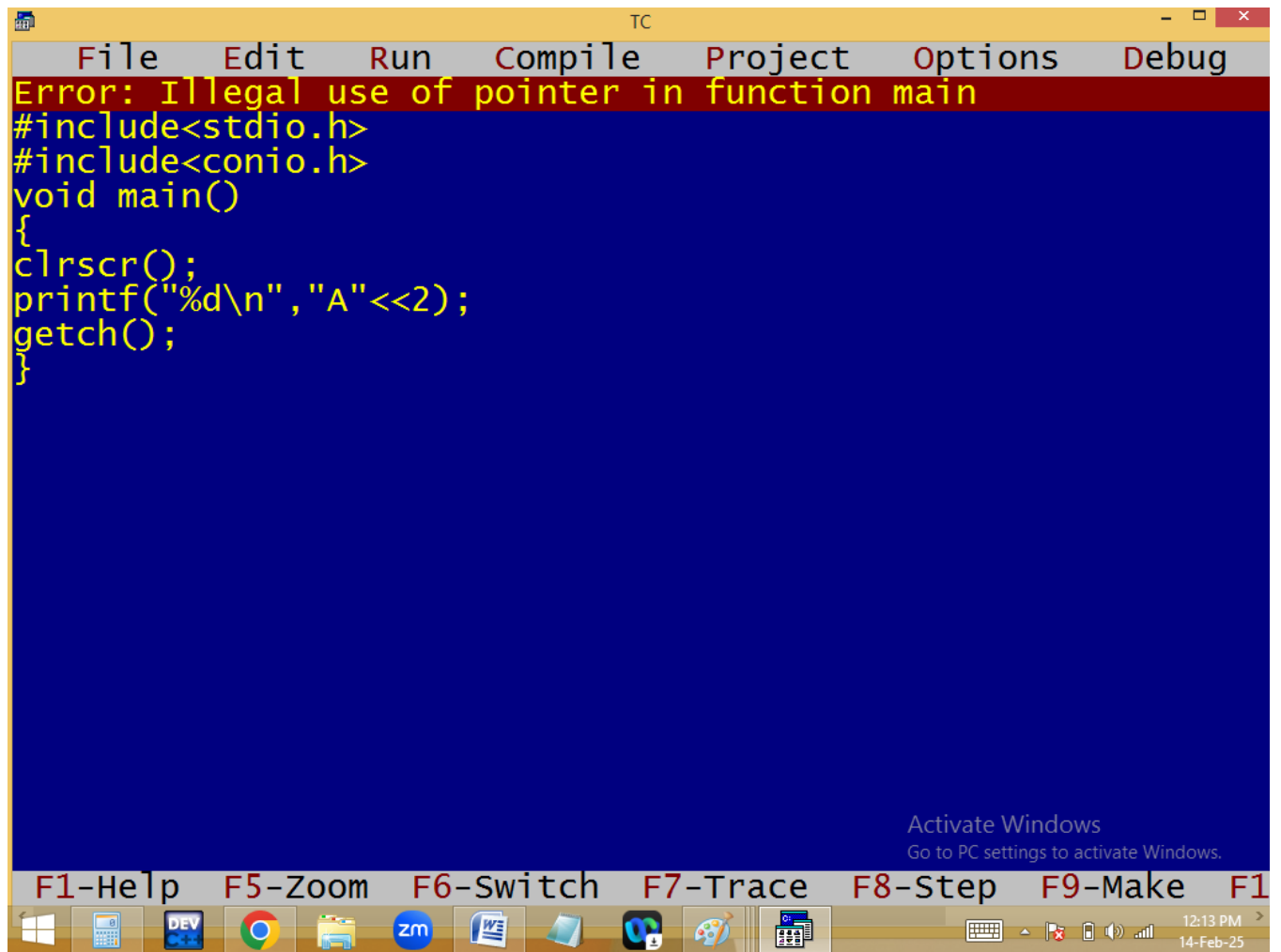


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. Below the menu bar, a status bar shows "Line 6", "Col 18", and various editing options like Insert, Indent, Tab, Fill, Unindent, and *. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 'A'<<2);
getch();
}
```

At the bottom of the IDE window, there is a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. Below this, the Windows taskbar is visible with icons for Windows, Calculator, DEV, Chrome, File Explorer, Zoom, Notepad, and others. The system tray on the right shows the time as 12:13 PM and the date as 14-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

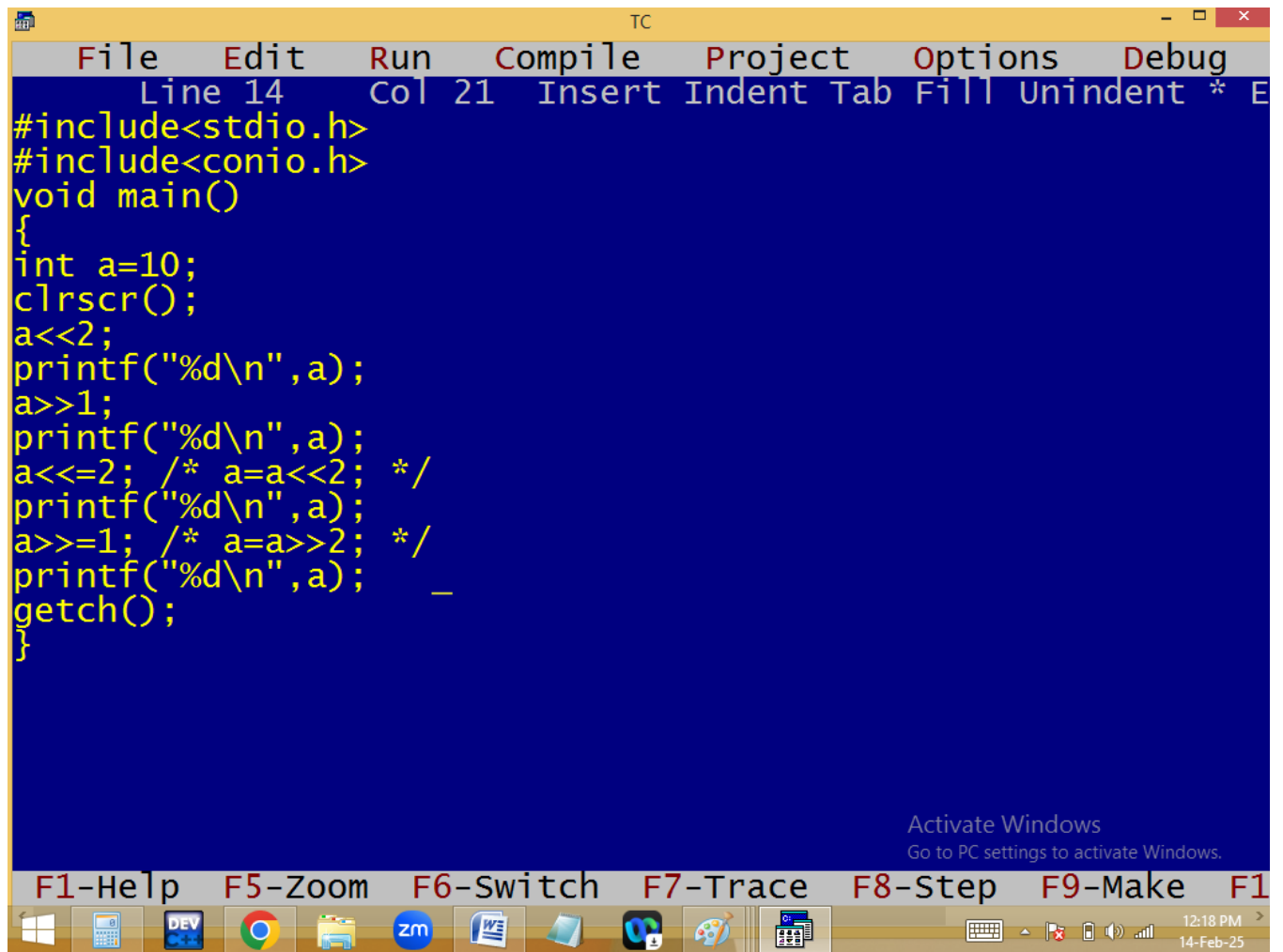




The screenshot shows the Turbo C++ (TC) IDE interface. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top states: "Error: Illegal use of pointer in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n","A"<<2);
getch();
}
```

At the bottom of the IDE window, there is a status bar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, and several applications (DEV, Chrome, File Explorer, Zoom, Word, PowerPoint, and a calculator). The system tray on the right shows the date and time: "12:13 PM" and "14-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the desktop area.

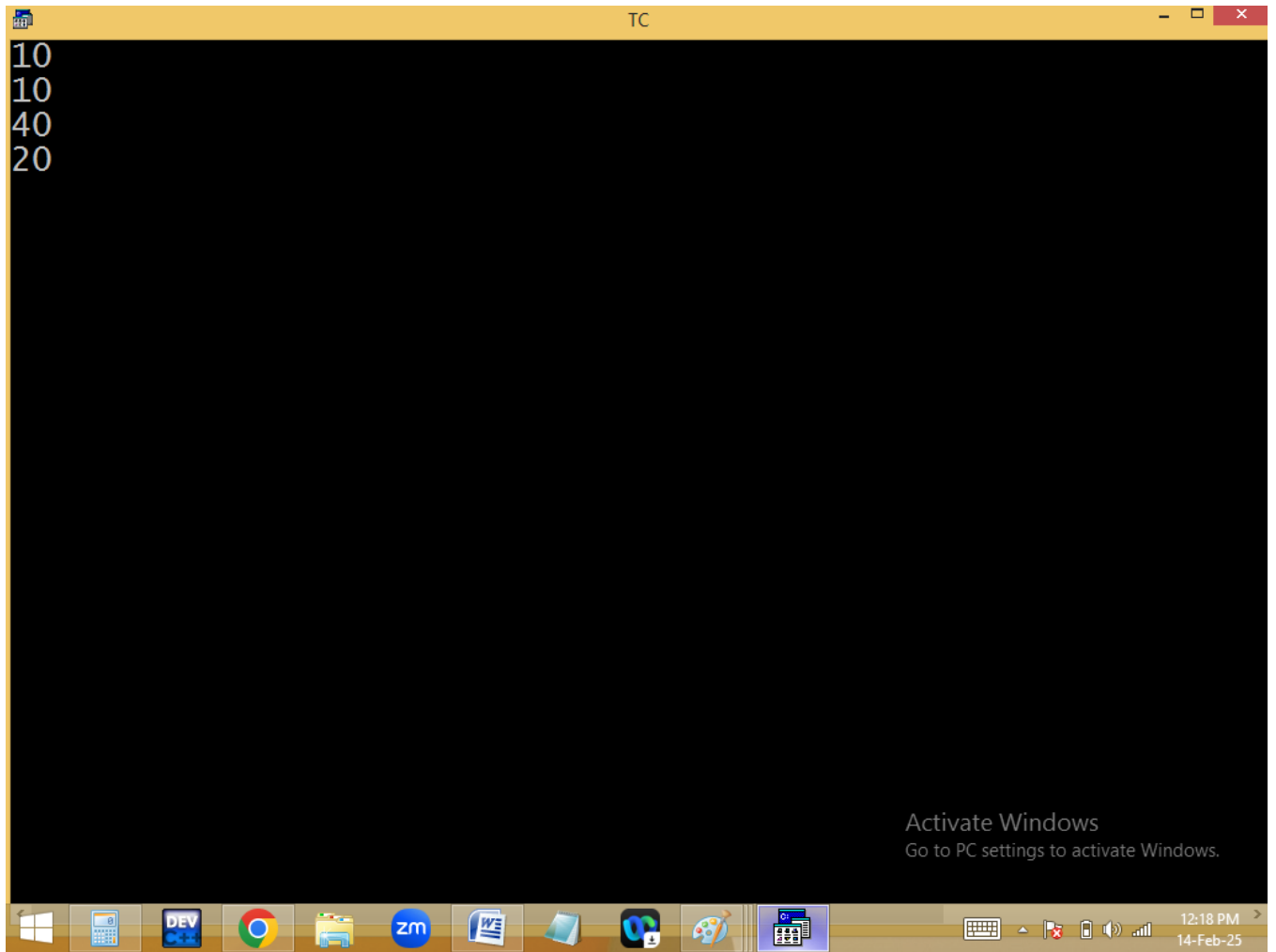


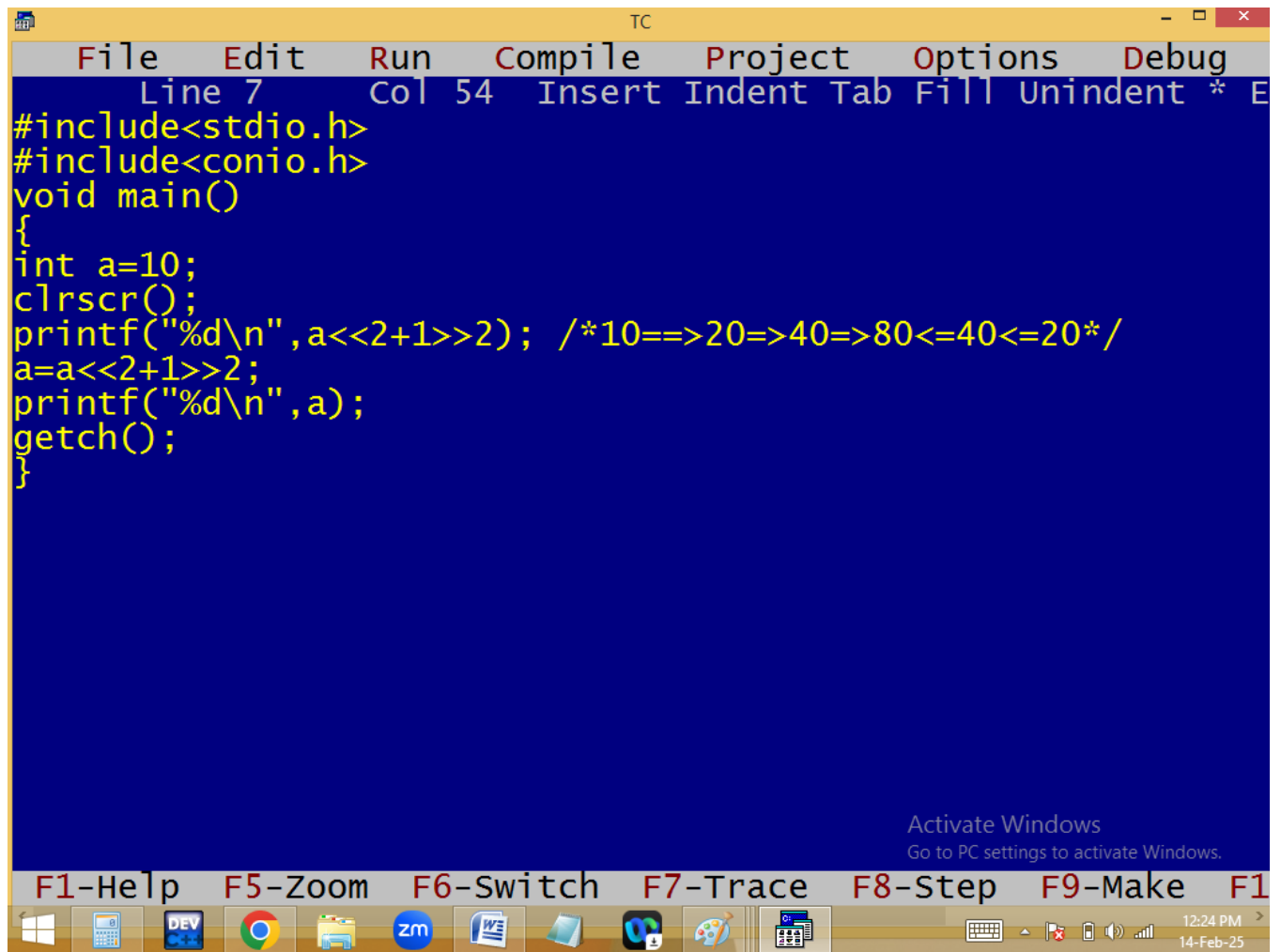
```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 21 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a<<2;
printf("%d\n",a);
a>>1;
printf("%d\n",a);
a<=<2; /* a=a<<2; */
printf("%d\n",a);
a>=>1; /* a=a>>2; */
printf("%d\n",a);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

12:18 PM
14-Feb-25



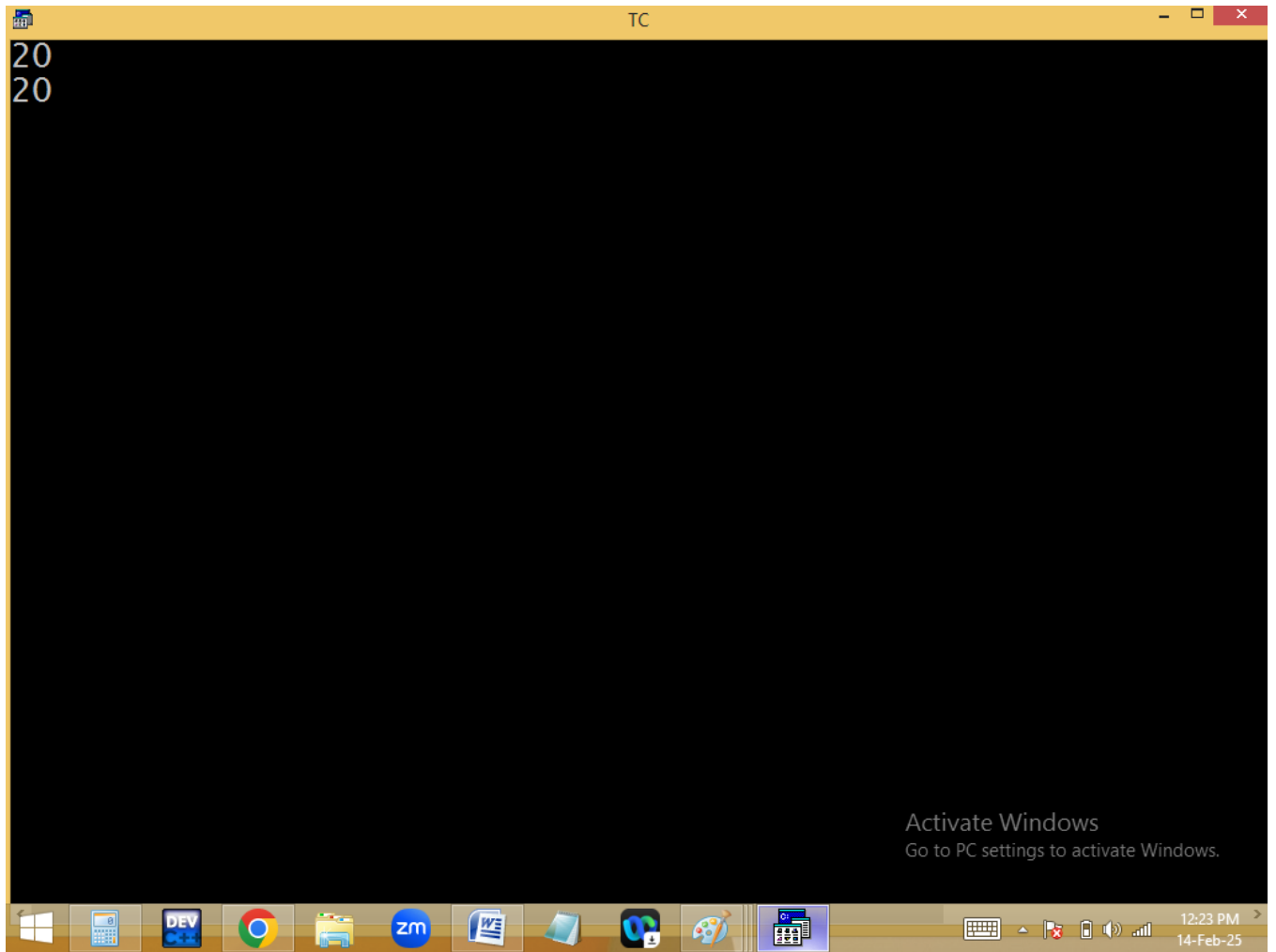


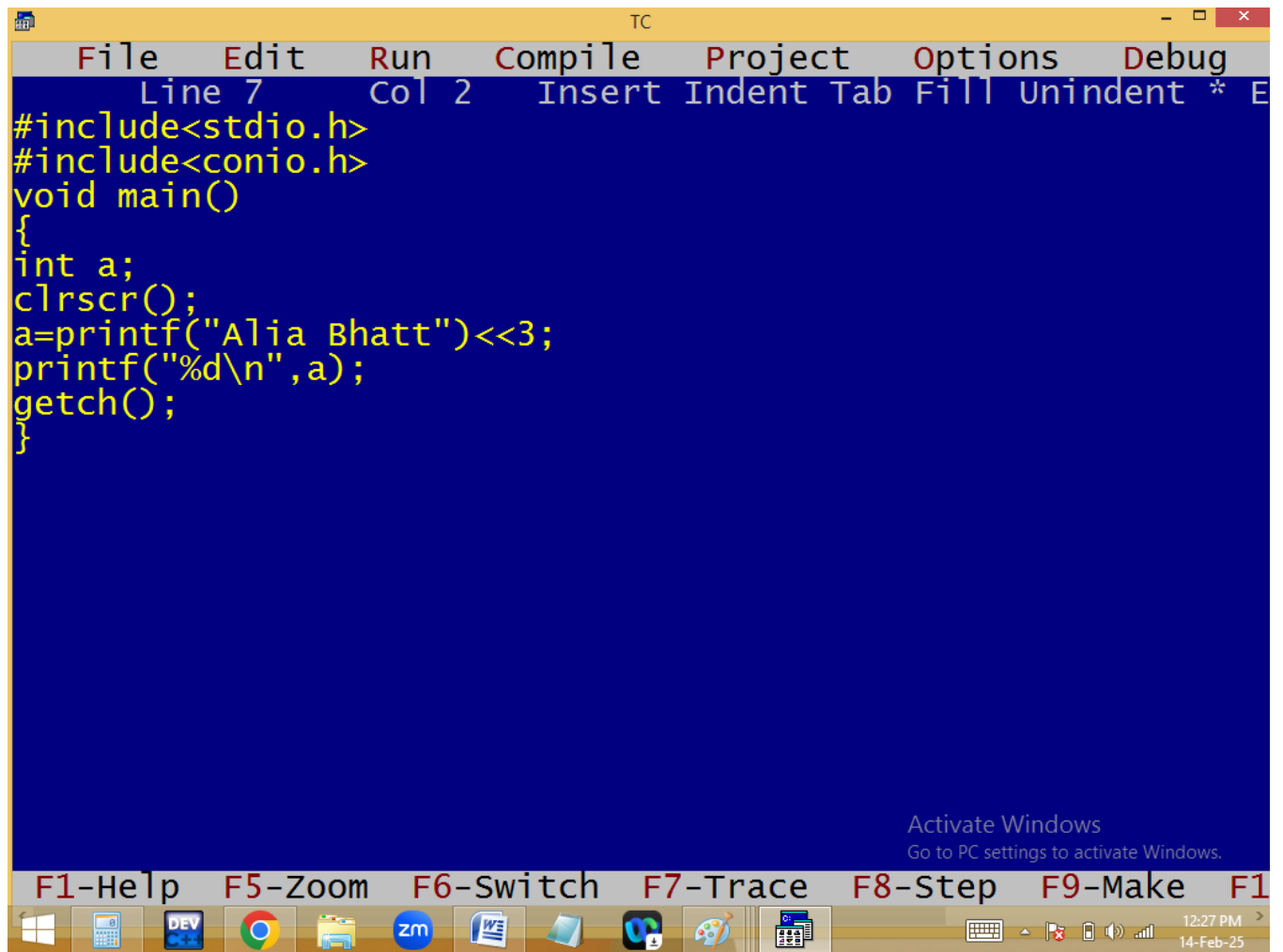
```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 54 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
printf("%d\n",a<<2+1>>2); /*10==>20=>40=>80<=40<=20*/
a=a<<2+1>>2;
printf("%d\n",a);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

12:24 PM
14-Feb-25



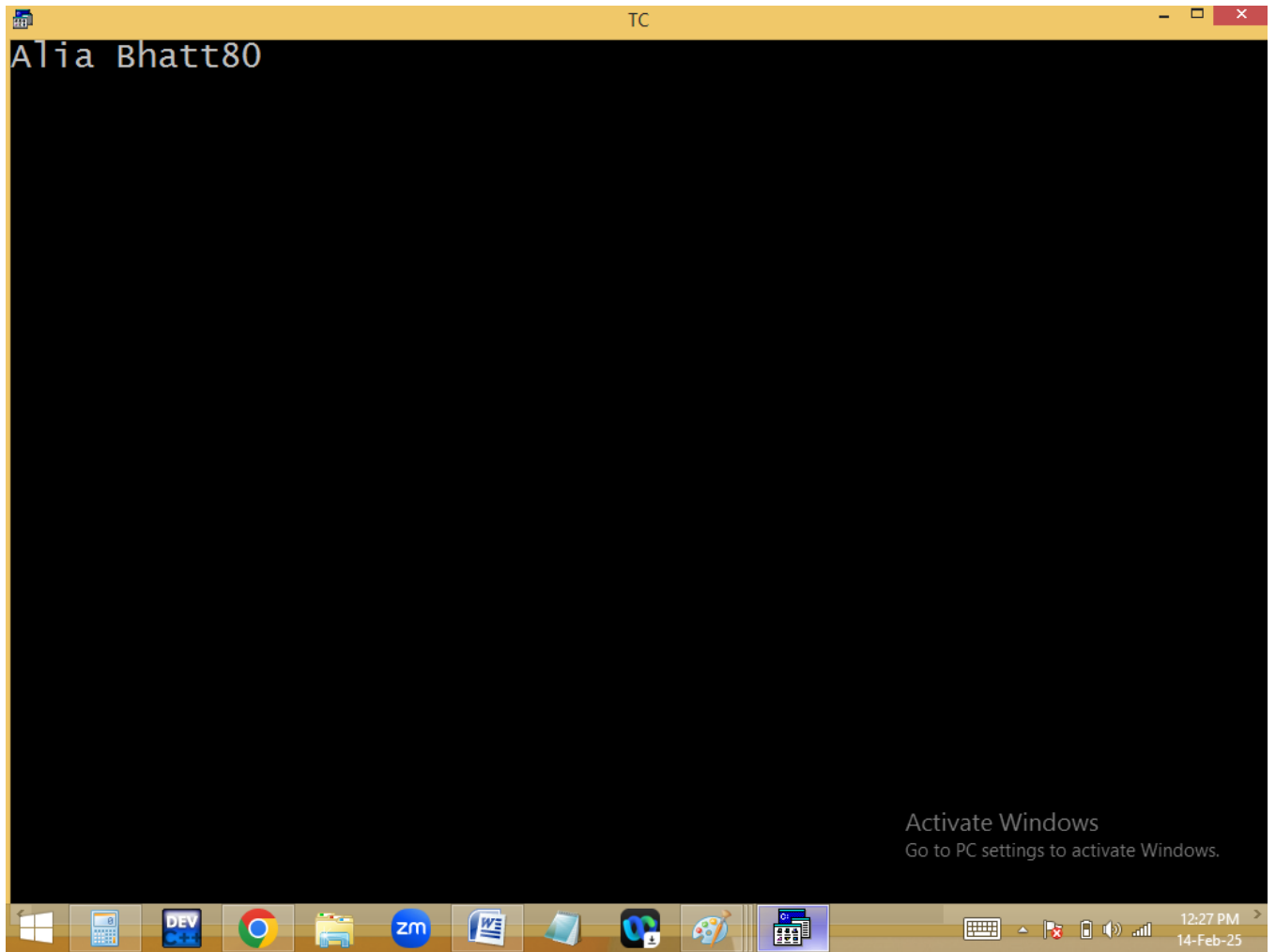


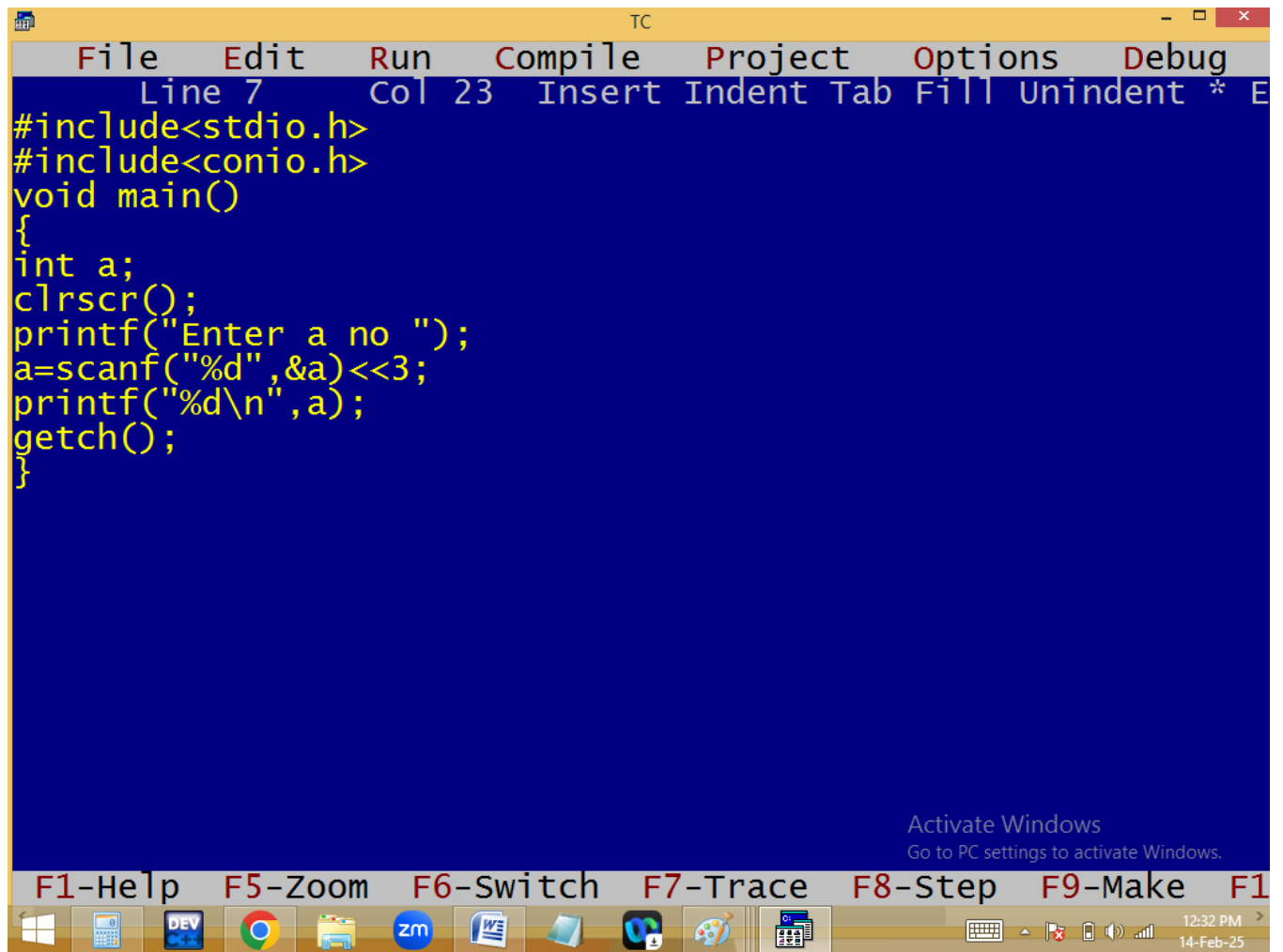
```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 2 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
a=printf("Alia Bhatt")<<3;
printf("%d\n",a);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

12:27 PM
14-Feb-25



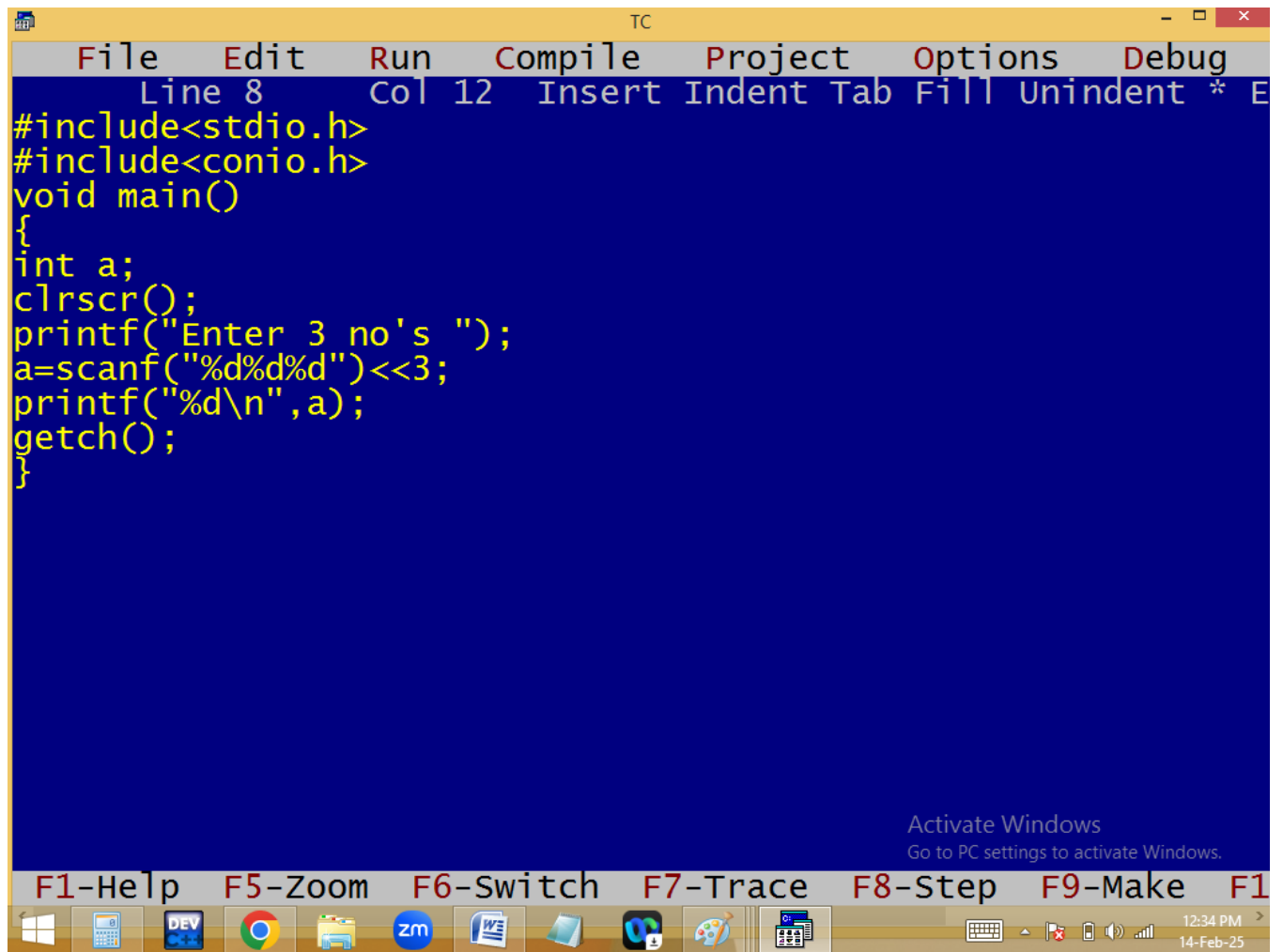


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 7", "Col 23", and lists editing actions: Insert, Indent, Tab, Fill, Unindent, and End of Line (* E). The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
printf("Enter a no ");
a=scanf("%d",&a)<<3;
printf("%d\n",a);
getch();
}
```

At the bottom of the IDE, there is a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. Below this is the Windows taskbar with icons for the Start menu, Task View, DEV, Chrome, File Explorer, Zoom, Notepad, and several other applications. The system tray on the right shows the time as 12:32 PM and the date as 14-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

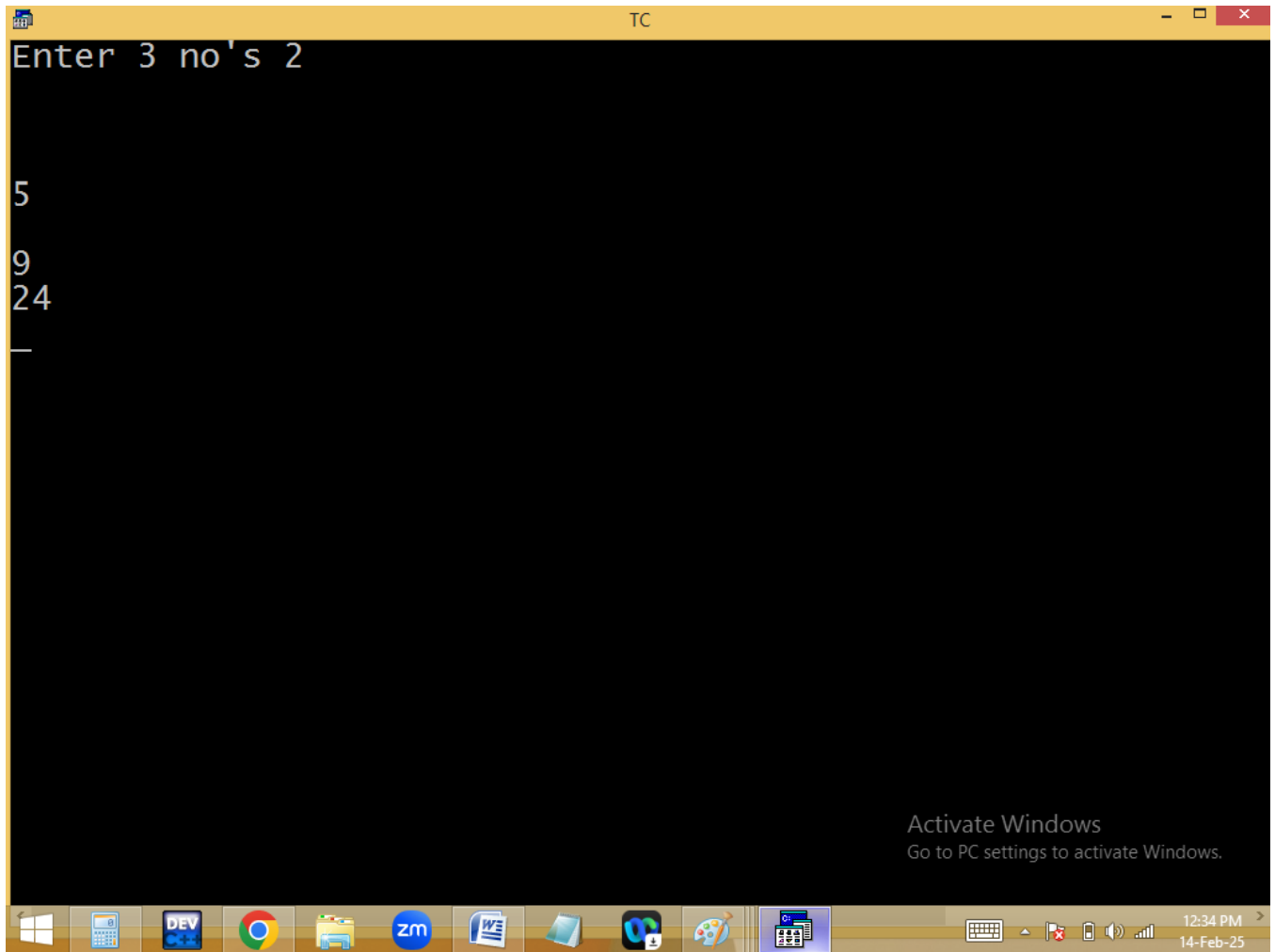




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8 Col 12 Insert Indent Tab Fill Unindent * E". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a;
    clrscr();
    printf("Enter 3 no's ");
    a=scanf("%d%d%d")<<3;
    printf("%d\n",a);
    getch();
}
```

At the bottom of the IDE, there is a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. Below this is the Windows taskbar with icons for Start, Task View, DEV, Chrome, File Explorer, Zoom, Notepad, and others. The system tray shows the time as 12:34 PM and the date as 14-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



The screenshot shows a Turbo C++ (TC) window with a yellow title bar. The main window has a black background with white text. The prompt "Enter 3 no's 2" is at the top. Below it, the input "5" is on the first line, "9" on the second line, and "24" on the third line. A horizontal line is on the fourth line. The Windows taskbar is at the bottom, showing various icons and the system clock.

```
Enter 3 no's 2

5
9
24
—

Activate Windows
Go to PC settings to activate Windows.
```

C programming structure:

[documentation section – comments – prog
heading s / meaning]

